

712.

A PARTIAL DIFFERENTIAL EQUATION CONNECTED  
WITH THE  
SIMPLEST CASE OF ABEL'S THEOREM.

[From the *Report of the British Association for the Advancement of  
Science*, (1881), pp. 534, 535.]

CONSIDER a given cubic curve cut by a line in the points  $(x, p)$ ,  $(x_1, p_1)$ ,  $(x_2, p_2)$ ; using the first and second points as elements, these determine uniquely the third point. Analytically, the equation of the curve determines  $p$  as a function of  $x$ , and  $p_2$  as a function of  $x_1$ ; writing in the equation

$$c_2 = \lambda c_1 + (1 - \lambda)c, \quad y_2 = \lambda y_1 + (1 - \lambda)y_0,$$

we have  $c$ , by a simple equation, and thence  $p_2$ ; viz.  $x_2$  is found as a function of  $x$ ,  $x_1$ , and of the nine constants of the equation. Hence having the derived equations (in regard to  $x$ ,  $x_1$ ) of the first, second, and third orders, we have  $(1 + 2 + 3 + 4) = 10$  equations from which to eliminate the 6 quantities  $x_2$ , considered as a function of  $x$  and  $x_1$ , thus involves a partial differential equation of the third order, independent of the particular cubic curve.

To obtain this equation it is only necessary to observe that we have, by Abel's theorem,

$$\frac{dx}{X} + \frac{dx_1}{X_1} + \frac{dx_2}{X_2} = 0,$$

where  $X$ ,  $X_1$ ,  $X_2$  is a given function of  $x$ , and  $p$ , that is, of  $x$ ;  $X_1$  and  $X_2$  are the like functions of  $x_1$  and  $x_2$  respectively. Hence, considering  $x_2$  as a function of  $x$  and  $x_1$ , we have

$$\frac{dx_2}{dx} = -\frac{X_2}{X}, \quad \frac{dx_2}{dx_1} = -\frac{X_2}{X_1},$$

[C. XI. 4-2]

and consequently

$$\frac{dx_2}{dx_1} \div \frac{dx_2}{dx} = \frac{X}{X_1},$$

where  $X$ ,  $X_1$  are functions of  $x_2$ ,  $x_1$  respectively; hence taking the logarithm and differentiating successively with regard to  $x_1$  and  $x_2$ , we have

$$\frac{d}{dx_2} \frac{d}{dx_1} \log \left( \frac{dx_2}{dx_1} \div \frac{dx_2}{dx} \right) = 0,$$

which is the required partial differential equation of the third order.

This differential equation has a simple geometrical significance. Consider three consecutive positions of the line meeting the cubic curve in the points 1, 2, 3; 1', 2', 3'; 1'', 2'', 3'' respectively; *quæ* equation of the third order, the equation should in effect determine 3'' by means of the other points. And, in fact, the three positions of the line constitute a cubic curve; the nine points are thus the intersections of two cubic curves, or, say, they are an “annexed” of points; any eight of the points thus determine uniquely the ninth point.

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## 713.

### ADDITION TO MR ROWE'S MEMOIR ON ABEL'S THEOREM.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXII., Part III. (1881), pp. 715—758. Received May 27,—Read June 16, 1880.]

IN Abel's general theorem  $y$  is an irrational function of  $x$  determined by an equation  $\chi(y) = 0$ , or say  $\chi(x, y) = 0$ ; if the order  $n$  as regards  $y$ , and that is also by him that the sum of any number of the integrals considered may be reduced to a sum of  $\rho$  integrals; where  $\rho$  is a determinate number depending only on the form of

the equation  $\chi(x, y) = 0$ , and given in his equation (42), [(*Euvres Complètes*, (1881), t. I., p. 157]; viz. if, solving the equation so as to obtain from it developments of  $y$  in descending series of powers of  $x$ , we have:

$$\begin{array}{ll} v_{\sigma}\rho_{\nu}, & \text{series each of the form } y = x^{\sigma'_{\nu}} + \dots, \\ & v_{\sigma}\rho_2, \quad y = x^{\sigma'_2} + \dots, \\ & \vdots \quad \quad \quad \vdots \\ & v_{\sigma}\rho_n, \quad y = x^{\sigma'_n} + \dots, \end{array}$$

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<sup>0</sup>The second power of a term coefficient: this being a relation  $y = x^p + \dots$ , representing the  $y$ , different values of  $y$  obtained by giving to the radical  $x^p$  and to its  $p$  values, and the corresponding values in the radicals which enter into representing  $y$ , values of  $y$ . It is assumed that the equation is such that the meaning is to make each representing  $y$ ,  $y_2$ . In each of the equations the meaning of  $y$  is taken positive in the development of  $y$  in descending powers of  $x$ .

(so that  $\sigma = v_1\rho_1 + v_2\rho_2 + \cdots + v_n\rho_n$ ) then  $y$  is a determinate function of  $x_1, x_2, x_3, x_4, x_5, x_6, \dots, x_n$ , say  $\rho$ .

We have so expressed Abel's  $\rho$  in the following form, viz. assuming

$$y = \Sigma a_1 \alpha_1 \nu_1 + \Sigma a_2 \nu_2 \cdots + \Sigma a_n \nu_n - \frac{1}{2} \Sigma a - \frac{1}{2} n + 1,$$

or, what is the same thing, for  $\rho$  writing this  $\Sigma a_1$ ,

$$y = \Sigma a_1 \nu_1 + \Sigma a_2 \nu_2 + \Sigma a_3 \nu_3 + \cdots + \Sigma a_n \nu_n - \frac{1}{2} \Sigma a - \frac{1}{2} n + 1,$$

where in the first sum  $\nu_1$  have each of them the values  $1, 2, \dots, \rho_1$  subject to the condition  $i + \nu_1 \sigma'_1$ ; in each of the other sums  $\nu_i$  and  $\rho_n$  are interrelated by the suffix, which has the values  $1, 2$ , etc.

It is a leading result in Riemann's theory of the Abelian integrals that  $y$  is the deficiency (Geschlecht) of the curve represented by the equation  $\chi(x, y) = 0$ ; and it must consequently be demonstrable *a posteriori* that the foregoing expression for  $y$  is in fact = deficiency of curve  $\chi(x, y) = 0$ . I propose to verify this by means of the formulae given in my paper "On the Higher Singularities of a Plane Curve," *Quart. Math. Journ.*, vol. VII., (1866), pp. 212—222, [374].

It is necessary to distinguish between the values of  $\frac{m}{n}$  which are  $>$ ,  $=$ , and  $< 1$ ; and to fix the ideas I assume  $l = 7$ , and

$$\begin{aligned} & \frac{m_1}{n_1}, \frac{m_2}{n_2}, \frac{m_3}{n_3}, \text{ each } > 1, \\ & \frac{m_4}{n_4} = 1; \text{ say } m_4 = n_4 = n, \text{ and } n_4 = n; \\ & \frac{m_5}{n_5}, \frac{m_6}{n_6}, \frac{m_7}{n_7}, \text{ each } < 1, \end{aligned}$$

but it will be easily seen that the reasoning is quite general. And now  $\Sigma$  to denote a sum in regard to the first set of suffixes  $1, 2, 3$ , and  $\Sigma'$  to denote a sum in regard to the second set of suffixes  $5, 6, 7$ . The foregoing value of  $\rho$  is thus

$$y = \Sigma a \nu + 4 + \Sigma' a \nu.$$

Introducing a third coordinate  $z$  for homogeneity, the equation  $\chi(x, y) = 0$  of the curve will be

$$0 = [y^{p-(n-1)}, y^{n-2}x^a, \dots] (y, z)^{n-2} + (y^{n-2}, y^{n-3}x^a, \dots) (y, z)^{n-1} + \cdots + (y)$$

where it is to be observed that  $\chi(x, y) = 0$  is the product of  $\nu_i$ , different ovaries; these divide themselves into  $n_i$

groups, each a product of  $\rho_i$  orates; and in each such product the  $\rho_i$  coefficients  $A_i$  are in general the  $\rho_i$  values of a function containing a radical  $a^{1/\rho_i}$  and are thus different from each other; it is to when follows in effect assumed in each step that all the  $a_i$ , but that all the  $x_i$  coefficients, etc. are different from each other.

The remarks apply to the other factors. It applies in particular to the factor  $(y, z)^{r-1}$ , viz. it is assumed that the coefficients  $A$  in the radical  $z, y^{n-2}, \dots$  are all of them different from each other. These assumptions as to the leading coefficients really imply Abel's assumption that  $\frac{m_1}{n_1}, \frac{m_2}{n_2}$  are all of them fractions in their least terms, viz. that  $n = 1$ , since

*1 rotate however for convenience the equation will be considered, by giving to*

In the product of the several infinite series, the terms containing the lowest powers of  $z, \frac{1}{n_1}$  is a function in its lowest terms, viz. since  $n_1 = 1$ .

A curve has singularities (singular points) at infinity, that is, on the line  $z = 0$ ; viz. *First*, a singularity at  $(x = 0, x' = 0)$ , where the tangent is  $x' = 0$ , and which, writing for convenience  $x = 1$ , is denoted by the function

$$(z - \nu^{m/n} x^{1-1/n})^{n-1} = \dots$$

where observe that the expressed factor indicates  $r_n$  branches  $(z - \nu^{m/n} x^{1-1/n})^{r_n}$ , or say  $r_n(m_n - p_n)$  partial branches  $z = \nu^{m/n} \dots$ , that is,  $r_n(m_n - p_n)$  partial branches  $z = \nu^{m/n} \dots$ , will in all  $r_n(m_n - p_n) + \dots$ , where  $r_n(m_n - p_n)$  partial branches at the encompassed factors with the suffixes 1' and 4; that is, we can regard the encompassed factors with the suffixes 1' and 4, where in the equation

$$z - \nu^{m/n} + \dots = 0.$$

*Secondly*, a singularity at  $(z = 0, y = 0)$ , where the tangent is  $y = 0$ , and which, writing for convenience  $z = 1$ , is denoted by the function

$$(y - z^{m/n} x^{1-1/n})^{n-1} = \dots$$

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<sup>0\*</sup> This assumption is virtually made by Abel, (1), [c. p. 157], in considers

where observe that the expressed factor indicates  $r_5$  branches  $(y - z^{m/n}x^{1-1/n})^{r_5}$ , or say  $r_5(n_5 - m_5)$  partial branches  $y = z^{m/n} \dots$ , that is,  $r_5(n_5 - m_5)$  partial branches  $y = z^{m/n} \dots$ , with in all  $r_5n_5 + \dots$ , with  $r_5n_5 + r_6n_6 + r_7n_7$  partial branches  $y = z^{m/n} \dots$ , with the suffixes 5, 6, 7.

*Thirdly*, suppose at  $(x = 0, y = 0)$ , where the tangent is  $z = 0$ , and consequently  $r_4n_4 = 0$ , partial branches  $z = y^0 \dots +$  similar terms with respect to  $x, y, z$ .

It is convenient to assume  $r_4 = 2$ , so that

$$\begin{aligned} E &= \frac{1}{2}(K - 1)(K - 2) = z, \\ E' &= \frac{1}{2}(K' - 1)(K' - 2) \\ &= \frac{1}{2}\Sigma(n - 1). \end{aligned}$$

hence

$$E + e' = \frac{1}{2}(K - 1) + \frac{1}{2}\Sigma(n - 1).$$

Assuming that there are no other singularities, the deficiency

$$\rho = \frac{1}{2}(K - 1)(K - 2) - E - e'$$

is

$$= \frac{1}{2}(K - 1)(K - 2) - \frac{1}{2}KK + \frac{1}{2}\Sigma(n - 1).$$

This should be equal to the before-mentioned value of  $\rho$ ; viz. we ought to have

$$\frac{1}{2}(K - 1)(K - 2) - \frac{1}{2}KK + \frac{1}{2}\Sigma(n - 1) = \Sigma a\nu_1 + 4 + \Sigma' a\nu_2 - \Sigma a - \Sigma' a - 2,$$

or, as it will be convenient to write it,

$$M = K - \frac{3}{2}K + \frac{1}{2}\Sigma(n - 1) - 1 = \Sigma a\nu_1 + \Sigma' a\nu_2 - \Sigma' a - 2,$$

which is the equation which ought to be satisfied by the values of  $N$  and  $\Sigma(n - 1)$  calculated, according to the method of my paper, for the foregoing singularities of the curve.

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the equation  $z^n = y^n \sqrt[n]{f(x)}$  as denoting integers; but where it appears that the indices  $a, b$ , are to be considered of fixed, and as in the values of degrees with respect to  $x$ . He says that if these be assumed as integers in the expression of  $y$  in descending powers of  $x$  (the leading coefficient being  $z$  in regard to  $y$ ), the singular factor are in  $(x, y)$  in regard to  $y$  in regard to  $x$ .

We have as before:

$$K = \Sigma n + \Sigma' n + \theta n.$$

The terms  $\Sigma n + \Sigma' n + \theta n$ , written at length, is

$$\begin{aligned} &= n_1(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) + \cdots + n_2(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_3(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_4(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_5(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_6(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_7(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3). \end{aligned}$$



which is

$$= \Sigma \nu_1 n_2 + \Sigma' n_2 (2\rho_2 + 2'\rho_1) + \Sigma \Sigma' + \Sigma' \Sigma + \Sigma a_1 a_2 a_3,$$

We have moreover

$$\Sigma' \rho_1 = \Sigma a_1 \rho_1 + (\theta + \Sigma' a_1) \rho_1 = \Sigma a_1 + \Sigma a_2 + \Sigma' a_1 \rho_1,$$

each branch  $(z - y^{m/n} x^{1-1/n})^{n-1}$  gives  $a = m_1 \rho_1$ , and the value of  $\Sigma(\kappa - 1)$  for this singularity is

$$n_1(m_1 - n_1 - 1) + r_1 n_1(m_1 - 1) = r_1 n_1 \rho_1 - n_1 \rho_1.$$

which is

$$= \Sigma a \nu_1 - \Sigma a \rho_1.$$

each branch  $(y - z^{m/n} x^{1-1/n})^{n-1}$  gives  $\sigma = m_n \rho_n$ , and the value of  $\Sigma(\kappa - 1)$  for this singularity is

$$n_4(m_4 - n_4 - 1) + r_4 n_4(m_4 - 1) = r_4 n_4 \rho_4 - n_4 \rho_4,$$

which is

$$= \Sigma' \nu_2 - \Sigma' \nu_4 - \Sigma' \nu_1.$$

For each of the  $\theta$  singularities

$$(y - z^{m/n} x^{1-1/n})^{n-1}.$$

we have  $a = n$ , and the value of  $\Sigma(\kappa - 1)$  is  $r = \theta(\kappa - 1)$ ; this is  $= 0$  for the value  $\lambda = 1$ , which is physically attributed to  $\lambda$ .

According to the theory explained in my paper above referred to, the several singularities not together equivalent to a certain number  $K$  of  $\kappa^*$  nodes and cusps; viz. we have

$$K = \Sigma n - \Sigma a + \frac{1}{2} \Sigma(\kappa - 1).$$

This should be equal to the before-mentioned value of  $\rho$ ; viz. we ought to have

$$(K-1)(K-2) - 2(K + \Sigma n + \Sigma' n + \theta n) = 2\Sigma a \nu_1 + 4 + 2\Sigma a \nu_2 - 2\Sigma a - 2\Sigma' a - 4,$$

or to it will be convenient to write

$$M - K - \Sigma K + 1 + \Sigma(n - 1) = \Sigma a \nu_1 + \Sigma' a \nu_2 - \Sigma a - \Sigma' a - 2,$$

which is the equation which ought to be satisfied by the values of  $N$  and  $\Sigma(n - 1)$  calculated, according to the method of my paper, for the foregoing singularities of the curve.

We have as before:

$$K = \Sigma n + \Sigma' n + \theta n.$$

The term  $\Sigma n + \Sigma' n + \theta n$ , written at length, is

$$= \Sigma \nu_1 n_1 + \Sigma' \nu_2 n_2 + \theta n,$$

[C. XI. 5]

or, reducing,

$$M = (\Sigma m)^2 - \Sigma ma - \Sigma m\rho_1 - \Sigma m\rho_2 + (\Sigma' m)^2 - \Sigma' m_2 a - \Sigma' m_2 \rho_2 \\ + (\Sigma' m)^2 - \Sigma' m\rho_1 - \Sigma ma + 2\Sigma' m_2 a_1 + 2\Sigma' \Sigma a_1 a_2;$$

and it is to be shown that the two lines of this expression are in fact the values of  $M$  belonging to the singularities

$$\left(z - y^{m/n} x^{1-1/n}\right)^{n-1}, \dots, \text{ and } \left(y - z^{m/n} x^{1-1/n}\right)^{n-1}, \dots,$$

respectively. We assume  $\lambda = 1$ , and there is then no singularity  $(y - z)^{n-1}$ .

I recall that, considering the several partial branches which meet in a singular point,  $M$  denotes the sum of the number of the intersections of each partial branch by every other partial branch: so that for each pair of partial branches the intersections are to be counted twice. Supposing that the tangent is  $x = 0$ , and that, for any two branches we have  $z = Az^x + \dots$  ( $x$  where  $p_x, p_x$  are each up to greater than 1), then if  $p_x = p_x$ , and  $A_x = A_x, A_x = A_x$  where  $A_x, A_x$  are not the partial branches which have here always could be regards the cases about to be considered), then the number of intersections is taken to be  $-$ ; and if  $p_x > p_x$ , and the number of intersections (taken to be  $\equiv p_x$ ) viz. in the case of as a unequal exponents, it is equal to the smaller exponents.

Consider now the singularity  $\left(z - y^{m/n} x^{1-1/n}\right)^{n-1}$ , and first the intersections of a partial branch  $z = y^{m/n} \dots$  by each of the remaining  $n_1(m_1 - p_1) - 1$  partial branches of the same set: the number of intersections with any one of these is  $= \frac{m_1}{n_1} [n_1(m_1 - p_1) - 1]$ . But we obtain this same number from each of the  $n_1(m_1 - p_1)$  partial branches, and thus the whole number is

$$n_1(m_1 - p_1) \frac{m_1}{n_1} [n_1(m_1 - p_1) - 1] = n_1 m_1 [n_1(m_1 - p_1) - 1].$$

Taking account of the other sets, and the whole number of intersections is

$$= \Sigma n^2 a - \Sigma' n a^2.$$

Observe now that  $\frac{m_1}{n_1} > \frac{m_2}{n_2}$ , that is,  $\frac{n_1}{n_2} < \frac{m_1}{m_2}$ , and that, these being each  $< 1$ , we have  $1 - \frac{m_2}{n_2} > 1 - \frac{m_1}{n_1}$ , that is,  $\frac{n_2 - m_2}{n_2} > \frac{n_1 - m_1}{n_1}$ , and we thus have

$$\frac{m_1}{n_1} < \frac{m_2}{n_2}, \text{ and } \left( z - y^{m/n} x^{1-1/n} \right)^{n_1(m_1-p_1)},$$

Considering now the intersections of partial branches of the two sets

$$\left( z - y^{m/n} x^{1-1/n} \right)^{n_1(m_1-p_1)} \text{ and } \left( z - y^{m/n} x^{1-1/n} \right)^{n_2(m_2-p_2)},$$

respectively, a partial branch  $z - z^{m/n} \dots$  gives with each partial branch of the other set a number  $= \frac{m_2}{n_2}$ ; and in this way taking each partial branch of each set, the number is

$$n_1(m_1 - p_1) \cdot n_2(m_2 - p_2) \cdot \frac{m_2}{n_2} = n_2 m_2 n_1(m_1 - p_1),$$

and thus for all the sets the number is

$$= 2 \Sigma n_2 m_2 \nu_2 m_2,$$

where in the first sum the  $\Sigma'$  refers to each pair of values of the suffixes. But the intersections are to be taken twice; the number thus is

$$= 2 \Sigma m_1 m_2 \nu_1 \nu_2,$$

Adding the foregoing number

$$= \Sigma' n^2 a - \Sigma' n a^2;$$

the whole number for the singularity in question is

$$= (\Sigma m_1)^2 - \Sigma' m_2 \rho_2 - \Sigma' m_2 \rho_2.$$

Similarly for the singularity  $\left( y - z^{m/n} x^{1-1/n} \right)^{n-1} \dots$ , taking each set with itself, the number of intersections is

$$n_5 m_5 [n_5(m_5 - p_5) - 1] + n_6 m_6 [n_6(m_6 - p_6) - 1] + n_7 m_7 [n_7(m_7 - p_7) - 1],$$

which is

$$= \Sigma' m_2^2 a - \Sigma' m_2 \rho_2 - \Sigma m_1 a.$$

$$[ \hspace{10em} 5 - 2 ]$$

We have here  $\frac{m_5}{n_5} > \frac{m_6}{n_6}$ , each of these being  $< 1$ , we have  $1 - \frac{m_5}{n_5} < 1 - \frac{m_6}{n_6}$ , that is,  $\frac{n_5 - m_5}{n_5} < \frac{n_6 - m_6}{n_6}$ , and so

$$\frac{m_5}{n_5} > \frac{m_6}{n_6}, \frac{n_5 - m_5}{n_5} < \frac{n_6 - m_6}{n_6};$$

Hence considering the two sets

$$\left(y - z^{m/n} x^{1-1/n}\right)^{n_5(m_5-p_5)} \quad \text{and} \quad \left(y - z^{m/n} x^{1-1/n}\right)^{n_6(m_6-p_6)};$$

a partial branch of the first set given with a partial branch of the second set a number of intersections; and the number then obtained is

$$n_5(m_5 - p_5) \cdot n_6(m_6 - p_6) \cdot \frac{m_6}{n_6} = n_6 m_6 n_5(m_5 - p_5),$$

For all the sets the number is

$$n_5 m_5 [n_5(m_5 - p_5) + n_6(m_6 - p_6)] - n_5 n_6(m_6 - m_5)$$

or taking these from, the number is

$$2\Sigma' m_2 m_2 \nu_2 \nu_2.$$

where in the first sum the  $\Sigma'$  refers to each pair of suffixes. Adding the foregoing value

$$\Sigma' m^2 a - \Sigma' m_2 \rho_2 - \Sigma m_1 a,$$

the whole number for the singularity in question is

$$= (\Sigma' m_2)^2 - \Sigma m_1 a - \Sigma' m_2 \rho_2.$$

and the proof is thus completed.

Referring to the first-note (ante, p. 31), I remark that the theorem “ $\rho$ -deficiency, is absolute, and applies to a curve with any singularities whatever: in a curve which has singularities not taken account of in Abel’s theory, the “perhaps see particulars” which we have a give as he expresses it, taken account of, viz. for that  $\Sigma a$  is, in the value of  $\rho$  as denoted by Abel’s formula; and the actual deficiency will be = Abel’s  $\rho$ , = such diminution, that is, it will be = true values of  $\rho$ .

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## VARIOUS NOTES.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 69, 115, 124, 125.]

## AN IDENTITY.

THE following remarkable identity is given under a slightly different form by Gauss, *Werke*, t. III., p. 419.

$$\begin{aligned} 1 &= \left(\frac{1}{x}\right)^2 x + \left(\frac{1}{1+x}\right)^2 x + \left(\frac{1+x}{1+x+x^2}\right)^2 x + \dots \\ &= \left[1 + \left(\frac{1}{x}\right)^2 x + \left(\frac{1}{1+x}\right)^2 x + \left(\frac{1+x}{1+x+x^2}\right)^2 x + \dots\right]^2. \end{aligned}$$

## ON TWO RELATED QUADRIC FUNCTIONS.

Assume

$$\begin{aligned} \varphi x &= x^2(1-x) + x(1-x^2-\alpha x), \\ \varphi y &= y(1-x) + y(x^2-1-\alpha x). \end{aligned}$$

then

$$\begin{aligned} \varphi\left(\frac{x^2}{x-a}\right) &= \frac{x^2}{(x-a)^2} \varphi x, \\ \varphi\left(\frac{x^2}{x-a}\right) &= \frac{x^2}{(x-a)^2} \varphi y. \end{aligned}$$

In the like of those for a ratio  $\frac{p}{1-p}$ , then

$$\varphi\left[\frac{ax^2(x-y)}{x^2-y^2(x-a)}\right] = \frac{ax^2(x-y)^2}{(x-a)^2-y^2(x-a)^2} \varphi(x).$$

*A TRIGONOMETRICAL IDENTITY.*

$$\begin{aligned} & \cos(b-c) \sin(b+c-a) + \cos a \sin(b+c) \\ & + \cos(c-a) \sin(c+a+b) + \cos b \sin(c+a) \\ & + \cos(a-b) \sin(a+b+c) + \cos c \sin(a+b) \\ = & \cos a \cos b \cos(b+c) + \cos b \cos c \cos(c+a) + \cos c \cos a \cos(a+b) + \cos a \cos b \cos(b+c) \end{aligned}$$

*EXTRACT FROM A LETTER.*

“I wish to construct a correspondence such as

$$(x + iy)^2(x + iy) = X + iY,$$

or, say, for greater convenience

$$4(x + iy)^2(x + iy) = X + iY,$$

viz. if

$$x + iy = \cos \alpha,$$

then

$$X + iY = \cos \alpha \beta.$$

Suppose  $\alpha$ , in a value of  $\alpha$  corresponding to a given value of  $X + iY$ , then the three values of  $x + iy$  are of course  $\cos \alpha$ ,  $\cos(\alpha + \frac{2\pi}{3})$ , but I am afraid that the calculation of  $\alpha$ , even with red and card tables, would be very laborious. Writing

$$X + iY = R(\cos \Theta + i \sin \Theta),$$

the intervals for  $\Theta$  might be  $3^\circ$ ,  $30^\circ$  or even  $15^\circ$ ; those of  $R$ , say 0.1 from 0 to 2, and then 0.2 up to 4; and 2 places of decimals would be quite sufficient; but even this would probably involve a great mass of calculation.

It has occurred to me that perhaps a geometrical solution might be found for the equation  $X + iY = \cos \beta$ .”

*October 21, 1877.*

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715.

NOTE ON A SYSTEM OF ALGEBRAICAL  
EQUATIONS.[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 17, 18.]

ASSUME

$$\begin{aligned}x + y + z &= P, \\yz + zx + xy &= Q, \\xyz &= R.\end{aligned}$$

$$\begin{aligned}A &= (nx + y - z) - a^n(nx + P), \\B &= y(ax + Q) - b^n(my + Q), \\C &= z(ax + Q) - c^n(nz + P), \\\Theta &= -mnQ + nP.\end{aligned}$$

Then

$$\begin{aligned}(ax + P)B - (my + P)A &= (ax + P)y(ax + Q) - a^n(my + Q) \\&= (ax + P)(ax + Q) - (my + Q)(my + P) \\&= ax(my + Q - ay) - yz(my + Q - P) + yz(ax + P) \\&= mny(ax - y) - P(ax - y) \\&= (ax - y)(mny - P) = (ax - y)\Theta,\end{aligned}$$

whence, identically,

$$\begin{aligned}(ax + P)B - (my + P)A &= (ay - y)\Theta, \\(my + P)C - (nx + P)B &= (by - z)\Theta, \\(ax + P)A - (nx + P)C &= (cz - x)\Theta.\end{aligned}$$

Hence any two of the equations  $A = 0$ ,  $B = 0$ ,  $C = 0$  imply the third equation.



We have

$$\begin{aligned} A &= x \left[ (n + 1)x + y + z \right] - a^n \left[ (n + 1)x + y + z \right] \\ &= (x - a^n) \left[ (n + 1)x + y + z \right]; \end{aligned}$$

and similarly for  $B$  and  $C$ . The three equations therefore are

$$\begin{aligned} \frac{x}{x^n - a^n} &= \frac{y + z}{(n + 1)x^n + (n - 1)a^n}, \\ \frac{y}{y^n - b^n} &= \frac{x + z}{(n + 1)y^n + (n - 1)b^n}, \\ \frac{z}{z^n - c^n} &= \frac{x + y}{(n + 1)z^n + (n - 1)c^n}, \end{aligned}$$

and any two of these equations imply the third equation.

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## 716.

AN ILLUSTRATION OF THE THEORY OF THE  
 $\vartheta$ -FUNCTIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 27—32.]

IF  $X$  be a given quartic function of  $x$ , and if  $a$ , as for convenience a constant multiple as, be the value of the integral  $\int \frac{dx}{\sqrt{X}}$ , taken from a given inferior limit to the superior limit  $x$ ; then, conversely,  $x$  is expressible as a function of  $a$ , viz. it is expressible in terms of  $\vartheta$ -functions of  $a$ , where  $\vartheta a$ , or say  $\vartheta(a, \Omega)$ , is the definition of which, depending, depend in general by definition on the sum of a series of exponentials of  $a$ , and it is possible from the assumed equation  $a = \int \frac{dx}{\sqrt{X}}$ , and the definition of  $\vartheta a$ , to attain by general theory the actual formulae for the determination of  $a$  as such a function of  $a$ .

I propose here to obtain these formula, in the case where  $X$  is a product of real factors, in a less suitable manner, viz. by reducing the integral  $\int \frac{dx}{\sqrt{X}}$  by a linear substitution in the form of an elliptic integral, the object being to obtain for the same in question the actual formulae as a function of  $\vartheta$ -functions of  $u$ .

The definition of  $\vartheta u$ , when the parameter is expressed,  $\vartheta(u, \Omega)$  is

$$\vartheta u = 2\Sigma(-)^p q^{p+\frac{1}{2}(p+1)^2} \sin(p + \frac{1}{2})u,$$

where  $u$  and  $p$  are negative integer values, zero included, from  $-\infty$  to  $+\infty$  (that is, from  $-2$  to  $2$ ,  $0$  to  $2$ , etc.); the parameter  $\Omega$ , or  $\Omega'$  (of imaginary) no real part, must be positive.

[C. XI.

6]

Evidently  $\vartheta u$  is an even function:  $\vartheta(-u) = \vartheta u$ . Moreover, it is at once seen that we have

$$\vartheta(u + \pi) = \vartheta u, \quad \vartheta(u + i\Pi) = -q^{-1}e^{-iu}\vartheta u,$$

where  $q$  is

$$\vartheta(u + \pi + i\Pi).$$

where as  $u$  and  $v$  are positive or negative integers, the product of  $\vartheta u$  by an exponential factor, or say simply that it is a multiple of  $\vartheta u$ .

Writing  $u = \frac{\pi}{2}\xi$ , we have  $\vartheta v = \frac{1}{2}\sqrt{2K}\vartheta\xi$ , that is,

$$\vartheta\left(\frac{\pi}{2}\xi\right) = \xi^{nK},$$

and therefore also

$$\vartheta(\sin \xi + v) = \xi^{nK}\vartheta v.$$

The above properties are general, but if  $\theta$  be real, then  $K, K', q$  being as in Jacobi (consequently  $q$  being real, positive, and less than 1, and  $K$  and  $K'$  real and positive), and assuming  $\Omega = \frac{K'}{K}, \pi$ , we have in the same thing,

$$q(= e^{-\pi\Omega/\Omega}) = e^{-\pi K'/K},$$

the function  $\vartheta v$  is given (*ut ante*) of Jacobi's  $\vartheta v$  by the equation  $\vartheta u = q^{\frac{1}{4}}\vartheta x$ .

That is, if  $\Omega$  is the value of the case  $u = v$ .

We have, as before, the system of three equations

$$\operatorname{sn} u = \frac{1}{\sqrt{k}} \frac{\vartheta(u, k)}{\vartheta_0(u, k)}, \quad \operatorname{cn} u = \sqrt{\frac{k'}{k}} \frac{\vartheta_0(u, k)}{\vartheta_0(u, k)}, \quad \operatorname{dn} u = \sqrt{k'} \frac{\vartheta_0(u, k)}{\vartheta_0(u, k)}.$$

Consider now the integral

$$\int_a^b \frac{dx}{\sqrt{(x - \alpha, x - \beta, x - \gamma, x - \delta)}} = \int_a^b \frac{dx}{\sqrt{X}} \text{ suppose,}$$

where  $\alpha, \beta, \gamma, \delta$  are taken to be real, and the four factors regarded as fixed magnitudes, viz. it is assumed that  $b - a$  is the value of the radical at the inferior limit is positive, and as  $b$  is between  $\beta$  and  $\delta$ ,  $X$  is positive, and *a fortiori* then  $\sqrt{X}$  denotes the positive value of the radical; but if  $\alpha$  is between  $\beta$  and  $\delta$ , then  $X$  is negative, and we must

that the sign of  $\sqrt{X}$  is taken so that  $\sqrt{X}$  is equal to a positive multiple of  $i$ , and this being so the integral is taken from the inferior limit to the superior limit  $x$ , which we may.

Take a linear function of  $y$ , such that for

$$x = a, b, c, d,$$

$$y = 0, 1, \frac{1}{k^2}, \infty, \text{ respectively,}$$

so that,  $x$  increasing continuously from  $a$  to  $d$ ,  $y$  will increase continuously from 0 to  $\infty$ . We have

$$y = \frac{c-a}{c-b} \cdot \frac{x-b}{x-a},$$

$$y = \frac{c-a}{c-d} \cdot \frac{x-d}{x-a},$$

$$1-y = \frac{b-c}{c-b} \cdot \frac{c-a}{c-a},$$

$$1-k'y = \frac{b-c}{c-b} \cdot \frac{c-a}{c-a},$$

and, thence,

$$\sqrt{y \cdot 1 - y \cdot 1 - k'y} = \frac{(a-c)\sqrt{X}}{(c-a)\sqrt{(x-a) \cdot (x-a)}},$$

where  $\sqrt{\frac{(c-d)}{(c-d)}}$  is taken to be positive, and the signs of  $\sqrt{X}$  is fixed as above. For  $y$  between 0 and 1,  $y = \frac{1}{x^2}$ ,  $1-y$ ,  $1-k'y$  will be positive, and  $\sqrt{y}$  being between 1 and  $\frac{1}{x^2}$ ,  $1-y$  will be negative, but  $1-k'y$  will be positive, viz. the radical is then taken as a positive multiple of  $i$ .

We have moreover

$$dy = \frac{c-a}{c-b} \cdot \frac{dx}{(x-a)^2},$$

and therefore

$$\frac{dy}{\sqrt{y \cdot 1 - y \cdot 1 - k'y}} = \sqrt{(c-d) \cdot (c-a)} \cdot \frac{dx}{\sqrt{X}},$$

where  $\sqrt{(c-d) \cdot (c-a)}$  is positive; or, say,

$$\int \frac{dy}{\sqrt{y \cdot 1 - y \cdot 1 - k'y}} = \sqrt{(c-d) \cdot (c-a)} \cdot \int \frac{dx}{\sqrt{X}}.$$

[ 6 - 2 ]

Hence, writing  $y = \sin^2 u$ , we have

$$\Pi u = \sqrt{(c-d) \cdot (c-a)} \int_a^b \frac{dx}{\sqrt{X}},$$

and it is to be further noticed that to

$$x = a, b, c, d,$$

correspond

$$\Pi u = 0, \frac{1}{2}K, \frac{1}{2}K + \frac{1}{2}K'.$$

or we may say

$$u = 0, K, K + iK', 2K + iK'.$$

Writing for shortness

$$\frac{1}{\Pi} = \sqrt{(c-d)(c-a)} = u,$$

we have

$$\sin u = \int_a^b \frac{dx}{\sqrt{X}},$$

and moreover

$$\sin(K + iK') = \int_a^b \frac{dx}{\sqrt{X}},$$

$$n(K + iK') = \int_b^c \frac{dx}{\sqrt{X}},$$

$$n(2K + iK') = \int_c^d \frac{dx}{\sqrt{X}},$$

or if for a moment we write

$$\int_a^b \frac{dx}{\sqrt{X}} = A, \text{ \&c.}$$

then these equations are

$$\begin{aligned} nK &= A, \\ n(K + iK') &= A + B, \\ n(2K + iK') &= A + B + C. \end{aligned}$$

Hence  $B + C = 2A = B$ , or  $A = B - C$ , or  $B - C + D = 0$ , that is,

$$\int_a^b \frac{dx}{\sqrt{X}} = \int_b^c \frac{dx}{\sqrt{X}},$$

where observe as before that  $x = a$  to  $x = b$ , or  $x = c$  to  $x = d$ ,  $X$  is positive, and the radical  $\sqrt{X}$  is taken to be positive.

We have also

$$\begin{aligned} nK &= A = \int_a^b \frac{dx}{\sqrt{X}}, \\ niK' &= C - B = \int_a^b \frac{dx}{\sqrt{X}}. \end{aligned}$$

when, as before, from  $b$  to  $c$ ,  $X$  is negative, and the sign of the radical is such that  $\sqrt{X}$  is a positive multiple of  $i$ ; the last formula may be more conveniently written

$$niK' = \int_b^c \frac{dx}{\sqrt{-X}},$$

where, here  $b$  to  $c$ ,  $-X$  is positive, and  $\sqrt{(-X)}$  is to be taken as positive.

Collecting the results, we have

$$\int_a^b \frac{dx}{\sqrt{X}} = \frac{nK}{\sqrt{(c-d)(c-a)}}, \quad b \leq a \leq b, \dots b \leq b \leq b \leq a,$$

and also

$$k = \frac{c-b}{a-b} \cdot \frac{d-a}{c-a},$$

and thence conversely

$$k' = \frac{a+d-b-c + (b-c-a+d) \sin^2 n}{a+b-c-d + (b-c+a-d) \sin^2 n},$$

or, what is the same thing,

$$\begin{aligned} nu^2 &= \frac{b-d \cdot x-a}{b-a \cdot x-d}, \\ nu^2 &= \frac{c-d \cdot x-a}{c-a \cdot x-d}, \\ nu^2 &= \frac{d-a \cdot x-b}{a-b \cdot x-d}, \\ nu^2 &= \frac{b-d \cdot x-a}{b-a \cdot x-d}, \end{aligned}$$

where, in place of the elliptic functions  $nu$  we substitute their  $\vartheta$ -values, it will be recollected that  $\Omega$ , the parameter of the  $\vartheta$ -functions, has the value

$$\frac{K'}{K}\pi = \left( = \frac{\Omega'}{\Omega} \pi \right) = \pi \frac{\int_a^b \frac{dx}{\sqrt{-X}}}{\int_a^b \frac{dx}{\sqrt{X}}},$$



and, so before,

$$K = \int_a^b \frac{dx}{\sqrt{X}},$$

Hence, finally,  $a, b, K, K', \Omega$  denoting given functions of  $a, b, c, d$ , as above

$$\int_a^b \frac{dx}{\sqrt{X}} = nu,$$

we have severally

$$\begin{aligned} \frac{b-d}{c-d} \cdot \frac{a-c}{c-a} &= \frac{1}{k^2} \frac{\vartheta_1^2 u}{\vartheta_2^2 u}, & \frac{\vartheta_3^2 u}{\vartheta_2^2 u}, \\ \frac{d-a}{d-b} \cdot \frac{b-c}{a-c} &= \frac{k'^2}{k} \frac{\vartheta_2^2 u}{\vartheta_3^2 u}, & \frac{\vartheta_1^2 u}{\vartheta_2^2 u} + \frac{1}{2} K', \\ \frac{c-a}{d-a} \cdot \frac{b-d}{b-c} &= k'^2 \frac{\vartheta_4^2 u}{\vartheta_3^2 u}, \end{aligned}$$

which are the formulae in question.

The problem is to obtain these (and this in the more general case where  $a, b, c, d$  have any given (real or imaginary) values directly from the assumed equation

$$\int_a^b \frac{dx}{\sqrt{X}} = nu,$$

and from the foregoing definition of the function  $\vartheta u$ .

It may be recalled that the function  $\vartheta u$  is a doubly infinite product

$$\vartheta u = \prod \prod \left[ 1 + \frac{u}{m\pi + (n + \frac{1}{2})i\Pi} \right]$$

as and a positive or negative integers from  $-\infty$  to  $+\infty$ ; I purposely omit all further explanations as to limits, or what is the same thing,

$$\vartheta \frac{\pi}{2K} = \prod \prod \left[ 1 + \frac{u}{2mK + (2n + 1)iK'} \right]$$

and consequently that, disregarding constant and exponential factors, the foregoing expressions of

$$\frac{b-d}{c-d} \cdot \frac{a-c}{c-a}, \quad \frac{d-a}{d-b} \cdot \frac{b-c}{a-c}, \quad \frac{c-a}{d-a} \cdot \frac{b-d}{b-c},$$

are the squares of the expressions

$$\frac{X}{Y}, \frac{Y}{Z}, \frac{Z}{W}, \quad \text{where } X, Y, Z, W \text{ are respectively of the form}$$

$$\begin{aligned} & n \prod \prod \left[ 1 + \frac{u}{(2n+1)iK'} \right], \quad \prod \prod \left[ 1 + \frac{u}{(2m+1)iK'} \right], \\ & \prod \prod \left[ 1 + \frac{u}{2mK + iK'} \right], \quad \prod \prod \left[ 1 + \frac{u}{(2m+1)K + iK'} \right], \end{aligned}$$

where ( $m, n = 2mK + 2niK', \&c.$ , the *smha* over the  $m$  or the  $n$  denotes that the  $2m$  or the  $2n$  as the case may be) is to be changed into  $2m+1$  or  $2n+1$ . But this is a transcription which has apparently no application to the  $\vartheta$ -functions of more than one variable.

717.

## ON THE TRIPLE THETA-FUNCTIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 48—50.]

AS a specimen of mathematical notation, viz. of the notation which appears to me the easiest to read, and also to print, I give the definition and demonstration of the fundamental properties of the triple theta-functions.

*Definition.*

$$\vartheta(U, V, W) = \Sigma \exp \Theta,$$

where

$$\Theta = (A, B, C, F, G, H)(l, m, n)^2 + (U, V, W)(l, m, n).$$

$l$ , denoting the sum in regard to all positive and negative integer values from  $-\infty$  to  $+\infty$  (zero included) of  $l, m, n$  respectively.

$\vartheta(U, V, W)$  is considered as a function of the arguments  $(U, V, W)$ , and it depends also on the parameters  $(A, B, C, F, G, H)$ .

*First Property.*  $\vartheta(U, V, W) = 0$ , for

$$U = \frac{1}{2}\pi i + (A, H, G)(\alpha, \beta, \gamma),$$

$$V = \frac{1}{2}\pi i + (H, B, F)(\alpha, \beta, \gamma),$$

$$W = \frac{1}{2}\pi i + (G, F, C)(\alpha, \beta, \gamma),$$

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta$  being any positive or negative integer numbers, such that as  $\alpha + \beta + \gamma = \text{odd number}$ .

*Demonstration.* It is only necessary to show that to each term of  $\vartheta$  there corresponds a second term, such that the inclusion of the two exponentials differ by an odd multiple of  $\pi i$ .

Taking  $l, m, n$  the integers which belong to the one term, those belonging to the other term are

$$-(l + a), -(m + \beta), -(n + \gamma).$$

(where observe that one at least of the numbers  $\alpha, \beta, \gamma$  being odd, this system of values is not in any case identical with  $l, m, n$ ). The two exponents are

$$\Theta_i = (A, B, C, F, G, H)(l, m, n)^2 + (U, V, W)(l, m, n),$$

and

$$\Theta'_i = (A, B, C, F, G, H)(l+a, m+\beta, n+\gamma)^2 - (U, V, W)(l+\alpha, m+\beta, n+\gamma),$$

viz. the value of  $\Theta'$  is

$$\begin{aligned} &= (A, B, C, F, G, H)(l, m, n)^2 + 2(A, H, G)(l\alpha, \beta, \gamma) + 2(H, B, F)(m\alpha, \beta, \\ &+ (A, B, C, F, G, H)(a, \beta, \gamma)^2 - (U, V, W)(l, m, n) - (U, V, W)(\alpha, \beta, \gamma), \end{aligned}$$

and we then have

$$\begin{aligned} \Theta' - \Theta &= -2(A, B, C, F, G, H)(l, m, n)^2 + 2(A, H, G)(\alpha, \beta, \gamma)l + 2(H, \\ &+ 2(G, F, C)(\alpha, \beta, \gamma)n + (A, B, C, F, G, H)(a, \beta, \gamma)^2 \\ &- 2(U, V, W)(l, m, n) - (U, V, W)(\alpha, \beta, \gamma). \end{aligned}$$

which proves the theorem.

As to the notation, remark that, after  $(A, B, C, F, G, H)$  has been written out in full, we may instead of  $((A, B, C, F, G, H)(\alpha, \beta, \gamma)^2$  write

$$(\alpha, \beta, \gamma)^2,$$

and that we may use the like abbreviation

$$(A, \dots)(\alpha, \beta, \gamma) \text{ for } (A, H, G)(\alpha, \beta, \gamma), \text{ \&c.},$$

$$(H, \dots)(\alpha, \beta, \gamma) \text{ for } (H, B, F)(\alpha, \beta, \gamma), \text{ \&c.},$$

$$(G, \dots)(\alpha, \beta, \gamma) \text{ for } (G, F, C)(\alpha, \beta, \gamma), \text{ \&c.}$$

These are not only abbreviations, but they make the formula actually clearer, as bringing them into a similar compact; and I accordingly use them in the demonstration which follows.

*Second Property.* If  $U, V, W$ , denote

$$\begin{aligned} U' &= \pi i + (A, H, G)(\alpha, \beta, \gamma), \\ V' &= \pi i + (H, B, F)(\alpha, \beta, \gamma), \\ W' &= \pi i + (G, F, C)(\alpha, \beta, \gamma), \end{aligned}$$

respectively, where  $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$  are any positive or negative integers (zero values admissible), then

$$\vartheta(U+u, V+v, W+w) = \exp \left[ -2(\alpha U + \beta V + \gamma W) - (\alpha, \beta, \gamma)^2 \right] \vartheta(U, V, W),$$

or, say

$$= \exp \left[ -(A, \dots)(\alpha, \beta, \gamma)^2 - 2(U, V, W)(\alpha, \beta, \gamma) - (\alpha, \beta, \gamma)^2 \right] \vartheta(U, V, W)$$

*Demonstration.* Writing  $\vartheta(U, V, W) = \Sigma \exp \Theta$ , then in the expression of  $\Theta$ , we may in place of  $l, m, n$  write  $-l - \alpha, -m - \beta, -n - \gamma$ ; we thus obtain

$$\begin{aligned} \Theta_l &= (A, \dots)(l, m, n)^2 + 2(A, \dots)(l, m, n)(\alpha, \beta, \gamma) + (A, \dots)(\alpha, \beta, \gamma)^2 \\ &\quad + (U, V, W)(l, m, n) + (U, V, W)(\alpha, \beta, \gamma), \end{aligned}$$

which is

$$\begin{aligned} &= (A, \dots)(l, m, n)^2 \\ &\quad + 2(\alpha, \beta, \gamma)(\alpha U + \beta V + \gamma W) - 2(A, \dots)(\alpha, \beta, \gamma) + \Sigma'(A, \dots)(\alpha, \beta, \gamma)^2 - \Sigma a^2(A, \dots)(\alpha, \beta, \gamma)^2 \\ &\quad - 2(\alpha, \beta, \gamma)^2 + 2\alpha, \beta, \gamma) + (A, \dots)(\alpha, \beta, \gamma)^2 - 2(A, \dots)(\alpha, \beta, \gamma) \\ &\quad + (U, V, W)(l, m, n) + (U, V, W)(\alpha, \beta, \gamma), \end{aligned}$$

which is

$$\begin{aligned} &= (A, \dots)(l, m, n)^2 \\ &\quad + \Sigma'(A, \dots)(\alpha, \beta, \gamma)^2 + \Sigma a^2(A, \dots)(\alpha, \beta, \gamma) + \Sigma \alpha \beta (A, \dots)(\alpha, \beta, \gamma)^2 \\ &\quad + (U, V, W)(l, m, n) + \Sigma(A, \dots)(\alpha, \beta, \gamma) \end{aligned}$$

Hence, rejecting the last line, which (as on can readily verify) has the exponential multiplied as the case may be,  $\vartheta(U, V, W)$  as  $\vartheta(U, V, W)$  multiplied by the factor

$$\exp \left[ -(A, \dots)(\alpha, \beta, \gamma)^2 - 2(U, V, W)(\alpha, \beta, \gamma) - (\alpha, \beta, \gamma)^2 \right],$$

which is the theorem in question.

In many cases a formula, which belongs to an indefinite number of  $k$  letters, is most easily intelligible when written out for three letters, but it is sometimes convenient to speak of the letters  $l$ ,  $m$ , ... or even the letter  $l$ , and to write out the formula accordingly.

[C. XI.

7]

In fact, writing herein  $\alpha = \gamma = 0$ , that is,  $a = \gamma = 0$ , the right-hand side becomes  $a^2$  and the arts on the left-hand side are  $a = B$ ,  $A = B$ ,  $a = A = I$ , which suppose any four sum the sum of which is  $= 0$ .

Writing in the last-mentioned equation  $\alpha, \beta, \gamma$  for the  $a$ 's of  $A, B, \gamma, \delta$  respectively, the equation becomes

$$\begin{aligned} P &= \beta P, \\ Q &= 1 - a^2 + \beta^2 + By^2, \\ R &= 1 - By^2 - 2\alpha + By^2, \\ S &= 1 - By^2. \end{aligned}$$

$L, M, N$  as the like functions  $\alpha, \beta, \gamma, \delta$  respectively,  $A, B, C, D$  being the same as before. And in fact

$$\begin{aligned} -LQ &= -y^2(\alpha^2 + \alpha^2) + \beta^2 + (\beta^2 - \alpha^2) + B(y^2 - x^2) + B(\beta - \alpha) + \dots, \\ &= \frac{1}{B} [\beta^2 - (\beta - \alpha)^2 + By^2 - By^2] + B(y^2 - x^2) + \dots, \\ &= \frac{1}{B} [\beta^2(x^2 - y^2) + B(y^2 - x^2) + B^2(y^2 - x^2)], \end{aligned}$$

that is,

$$\begin{aligned} -LPQ &= -y^2(\beta^2 - \alpha^2) + \beta^2 - (\beta - \alpha) + \dots, \\ &+ (B + \alpha - \beta)\beta(y^2 - x^2) + \beta - B^2(y^2 - x^2) + \dots, \end{aligned}$$

or omitting the factor  $\beta$ , that is,

$$-LPQ = -y^2(\beta^2 - \alpha^2) + \beta^2 - y^2(x^2 - \alpha^2) + \dots + \beta(y^2 - x^2) - By^2(x^2 - \alpha^2),$$

It is easy to verify the form of the orders 0, 1, 2, 3, and 4 in  $x, y, z, w$  separately destroy each other, for instance, for the terms of the order 4, the value of  $\Sigma$  is then

$$\begin{aligned} &= A^2 + \Sigma y^2(\beta^2 - \alpha^2) + y(x^2 - \alpha^2) + B^2(x^2 - \alpha^2) + (\Sigma \beta^2)(x^2 - \alpha^2) + \dots, \\ &\frac{1}{B} [\Sigma \beta a^2(x^2 + y^2) + \Sigma a^2 B^2(x^2 - y^2) + B^2 a^2(x^2 - y^2)], \end{aligned}$$

It is easy to verify that the terms of the orders 0, 1, 2, 3, and 4 in  $x, y, z, w$  separately destroy any sum sum of  $x$  which is

$$-LP^2 + yQ^2 + yP^2 + xR^2 + yS^2 + (LP + yQ + xR + yS + yT)$$

$+(D^2-1-LP^2)(yP^2+yP^2-yQ^2-yQ^2)+2(LP+yQ+xR+yS+yT)(LP+yQ+x$   
or, omitting the factor  $\beta$ , that is,

$$-(LP+yQ^2+xR+yS+yT)+\Sigma(LP+yQ+xR+yS+yT)(LP+yQ+xR+yS+yT)$$

The theorem in the original form was obtained by me as follows:  
using the elliptic coordinates  $p, q, r$ , such that

$$\frac{p^2}{a+p} + \frac{q^2}{b+p} + \frac{r^2}{c+p} = 1,$$



or, what is the same thing,

$$-Bp\rho^2 = a + p \cdot b + p \cdot c + p,$$

$$-y\eta y = b + q \cdot a + q \cdot c + q,$$

$$-z\zeta z = c + r \cdot a + r \cdot b + r,$$

where  $A, B$  denote  $b - c, c - a, a - b$  respectively; then, treating  $r$  as a constant, the coordinates  $x, y, z$  will belong to a point on the ellipsoid

$$\frac{x^2}{a+r} + \frac{y^2}{b+r} + \frac{z^2}{c+r} = 1,$$

and the differential equation of the right lines upon this surface is

$$\frac{dp}{\sqrt{a+p \cdot b+p \cdot c+p}} + \frac{dq}{\sqrt{a+q \cdot b+q \cdot c+q}} = 0.$$

Take  $a, b, c$ , the coordinates of a point on the surface, and  $p, q$ , the corresponding values of  $p, q$ , so that

$$-Bp_0\rho_0^2 = a + p_0 \cdot b + p_0 \cdot c + p_0,$$

$$-Aq_0\eta_0 = b + q_0 \cdot a + q_0 \cdot c + q_0,$$

$$-z\zeta_0 = c + r \cdot a + r \cdot b + r,$$

then the equation of the tangent plane at the point  $(a, b, c, p_0, q_0)$  is

$$\frac{xz_0}{a+r} + \frac{yy_0}{b+r} + \frac{zz_0}{c+r} = 1,$$

or, substituting for  $x, y_0, z_0$  their values, we have

$$= \beta p_0 \cdot ay + \cdot bxz + Az + \dots,$$

and consequently the equation of the tangent plane through  $(a, b, c, p_0, q_0)$  is the integral of the proposed differential equation.

Writing

$$mz^2 = A(a+p), \quad mz^2 = B(b+p), \quad dx^2 = C(c+p),$$

the values of  $A, B, C, S$  are determined; and, assuming for  $q, r, p$ , the like forms with the arguments  $a, a_0, a_0$ , the differential equation becomes  $dz + dz$ , having the

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integral  $u + x_0 + v = u_0$ ; while the foregoing integral equation, on reducing the constant coefficients contained therein, takes the form

$$\begin{aligned}
 -l^2xx + mz_0mz + nz_0mz_0 + nz_0nz + nz_0nz_0 + \dots &= 0, \\
 &= \frac{1}{2}lm \cdot mz_0mz_0, \\
 &= -l^2nz \cdot mz_0mz_0, \\
 &= \frac{nz_0}{B^2 - 1 - l^2} \cdot \frac{l^2}{l^2 - l^2},
 \end{aligned}$$

via. this equation holds good if  $a = u_0 + v$ , we have the theorem.

If, as above,  $a = u + v + v$ ,  $b = u - v + v$ ,  $c = u + v - v$ , where  $r$  is regarded as constant, then the three products  $\text{sn}u \text{nz} \text{nz}$ ,  $\text{cn}u \text{nz} \text{nz}$ ,  $\text{dn}u \text{nz} \text{nz}$ , and equally so the three products  $\text{sn}z \text{nz} \text{nz}$ ,  $\text{cn}z \text{nz} \text{nz}$ ,  $\text{dn}z \text{nz} \text{nz}$ , are expressed by means of those products with the  $u$ ,  $v$ ,  $w$  of the  $u$ , of the  $v$ , viz. for two surfaces it would be possible to obtain a real relation equation, which may be expressed in the squares of the  $u$ ,  $v$ ,  $w$  of the  $r$ , of the  $z$ , all of whose internal coefficients are nominal, viz. the such would be obtained, in fact, a similar relation in the like form by giving to the parameter  $r$  the advantage of being of a degree which is the half of the equation which involves only the squared functions.

Write  $a$ ,  $b$ ,  $c$  or for  $\text{sn}u$ ,  $\text{cn}u$ ,  $\text{dn}u$ , respectively; then, writing

$$\begin{aligned}
 A &= \sqrt{1 - y^2 \cdot 1 - l^2y^2}, & a &= \sqrt{1 - l^2 \cdot 1 - l^2x^2}, \\
 B &= \sqrt{1 - l^2 \cdot 1 - l^2y^2}, & b &= \sqrt{1 - x^2 \cdot 1 - l^2y^2}, \\
 C &= \sqrt{1 - l^2 \cdot 1 - l^2y^2}, & c &= \sqrt{1 - l^2 \cdot 1 - x^2 - y^2}, \\
 D &= 1 - l^2x^2y^2, & d &= 1 - l^2z^2x^2,
 \end{aligned}$$

we have

$$\text{sn}(z + v) = \frac{xA + ya}{D},$$

that is,

$$\frac{A + P}{D} = \frac{\sigma c}{\sigma a \cdot \sigma b},$$

and consequently

$$Dn = (A - x)(A + x),$$

$$PD = (P + x)(P - x);$$

whence

$$Dn = PD = (Aa - x)(Aa - x).$$